Optimizing maintenance dosing of emicizumab-kxwh as prophylaxis in hemophilia A: Dosing to product labeling while minimizing drug waste

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Plain language summary

Emicizumab-kxwh has changed how patients with hemophilia A prevent bleeding episodes. It is a subcutaneous injection, like insulin, given every 1-4 weeks. Emicizumab-kxwh comes in 4 vial sizes. Choosing the right combination of vials for each dose is important to minimize medication waste. A pharmacist can assist prescribers in choosing the best vial combination to deliver the right dose to each patient in a way that minimizes waste and reduces cost without negatively affecting effectiveness.

Implications for managed care pharmacy

Hemophilia is among the most expensive disease states to manage, from both a medical and pharmacy benefit perspective. Treatment with emicizumab-kxwh, owing to its robust efficacy and relative ease of use, is expanding prophylaxis to patients who were previously treated only on demand. This larger pool of patients receiving prophylaxis is driving higher drug costs. These costs can be lowered when a specialist pharmacist reviews every regimen for appropriateness of dose, frequency of administration, and vial combination.

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ABSTRACT

BACKGROUND: The introduction of the bispecific antibody emicizumab-kxwh has driven a paradigm shift in the management of patients with hemophilia A. Emicizumab-kxwh is gradually replacing factor VIII as the treatment of choice for prophylaxis. The requirement to dose emicizumab-kxwh by weight can drive waste when the appropriate combination of various strengths of single-use vials is not considered, whereas inappropriate rounding to full vials can lead to significantly subtherapeutic or supratherapeutic doses.

OBJECTIVE: To quantify waste reduction and impact to savings when a clinical program designed to optimize the emicizumab-kxwh dose, frequency, and vial combination is applied at the dispensing specialty pharmacy level.

METHODS: The electronic medical records of 360 patients receiving emicizumab-kxwh as maintenance therapy were retrospectively reviewed. Patients were included in the sample if they were male, were aged 18 to 89 years, were inhibitor negative, and had at least 1 new or renewal maintenance prescription in 2021. Three hundred seventy discrete regimens were identified as evaluated per the Regimen Optimization Algorithm - emicizumab-kxwh and included in the final sample. Data collected for each regimen included patient weight, emicizumab-kxwh dose and interval originally prescribed, every 7-day equivalent regimen (to the hundredth of a milligram), vial combination originally ordered, and resulting waste generated. For those regimens failing the algorithm, additional review of pharmacist follow-up and prescriber engagement was performed. When the prescriber adopted a pharmacist recommendation to adjust dose, frequency of administration and/or vial combination, impact to cost was calculated (Medicare Part B 2022 average sale price).

RESULTS: 48% (176/370) of reviewed regimens failed specialty pharmacist review because they drove a potentially subtherapeutic or supratherapeutic dose and/or resulted in the use of partial vials that required that the patient discard all emicizumab-kxwh over the prescribed amount with each dose. 112 (64%) of these failed regimens met full criteria for prescriber engagement to recommend a regimen adjustment to dose, frequency of

administration, and/or vial combination. These recommendations were adopted in 43% of cases (48/112), resulting in a cumulative savings of 600 mg per dose while also avoiding significant subtherapeutic or supratherapeutic dosing. When factoring in the frequency of administration at the patient level, cumulative annual savings to payers across these 48 accepted recommendations was $1,793,549.76.

CONCLUSIONS: These data illustrate that the application of an evidence-based Regimen Optimization Algorithm at the dispensing specialty pharmacy level can reduce waste and drive payer savings while avoiding significant subtherapeutic or supratherapeutic dosing.

Introduced in the 1970s, concentrated factor VIII (FVIII) replacement has since been the gold standard for the management of hemophilia.[1] In subsequent decades, the manufacture of FVIII has changed significantly. Science has evolved from the complex extraction of FVIII out of human plasma pools to mass production of recombinant copies in the laboratory. More recently, the recombinant FVIII molecule has been modified to increase yield through b-domain deletion and/or extend half-life through fusion with carrier agents such as polyethylene glycol or proteins like albumin or IgG1.[2] Unfortunately, these advances have not significantly lessened the treatment burden associated with requisite intravenous administration and repeat dosing to treat and prevent bleeding episodes. This model changed dramatically with the approval of the first bispecific antibody (emicizumab-kxwh) currently indicated for the prevention of bleeding episodes in patients with hemophilia A, regardless of inhibitor status.[3]

Patients with hemophilia A do not produce adequate quantities of clotting FVIII sufficient to drive the clotting cascade in formation of a stable blood clot. Emicizumab-kxwh is a recombinant, humanized, bispecific antibody (IgG4) capable of binding activated factor IX and factor X, restoring the cascade by providing a bridge over the missing FVIII and rendering its absence obsolete.[4] As an antibody, emicizumab-kxwh has additional therapeutic advantages over FVIII replacement, including a significantly longer half-life (26.9 ± 9.1 days [SD]) and acceptable subcutaneous bioavailability (80.4%-93.1%).[5]

Emicizumab-kxwh was originally approved with labeling in support of the single dosing regimen 1.5 mg/kg by subcutaneous injection every 7 days.[6] Early population models validate that emicizumab-kxwh exhibits first-order pharmacokinetics following a 4-week loading regimen.[7] This understanding formed the hypothesis for a second phase 3 trial (HAVEN 2) evaluating 2 additional regimens: 3 mg/kg every 14 days and 6 mg/kg every 28 days.[8] In 2018, the US Food and Drug Administration (FDA) granted approval to update the product labeling in support of all 3 options.[9]

Emicizumab-kxwh is commercially available in 4 strengths: 30 mg/mL, 60 mg per 0.4 mL (equivalent to 150 mg/mL), 105 mg per 0.7 mL (equivalent to 150 mg/mL), and 150 mg/mL; all are single-use vials. Vials with the same concentration can be combined in the same syringe to decrease injection burden. The recommended maximum volume per injection is 2 mL. Volumes in excess of 2 mL should be separated into multiple injections.[5] A review of doses to possible vial combinations reveal that once a single dose reaches 90 mg, the amount of drug dispensed over the target dose and discarded should never need exceed 14 mg. Any drug waste in excess of 14 mg can be reduced or eliminated by adjusting the vial combination.

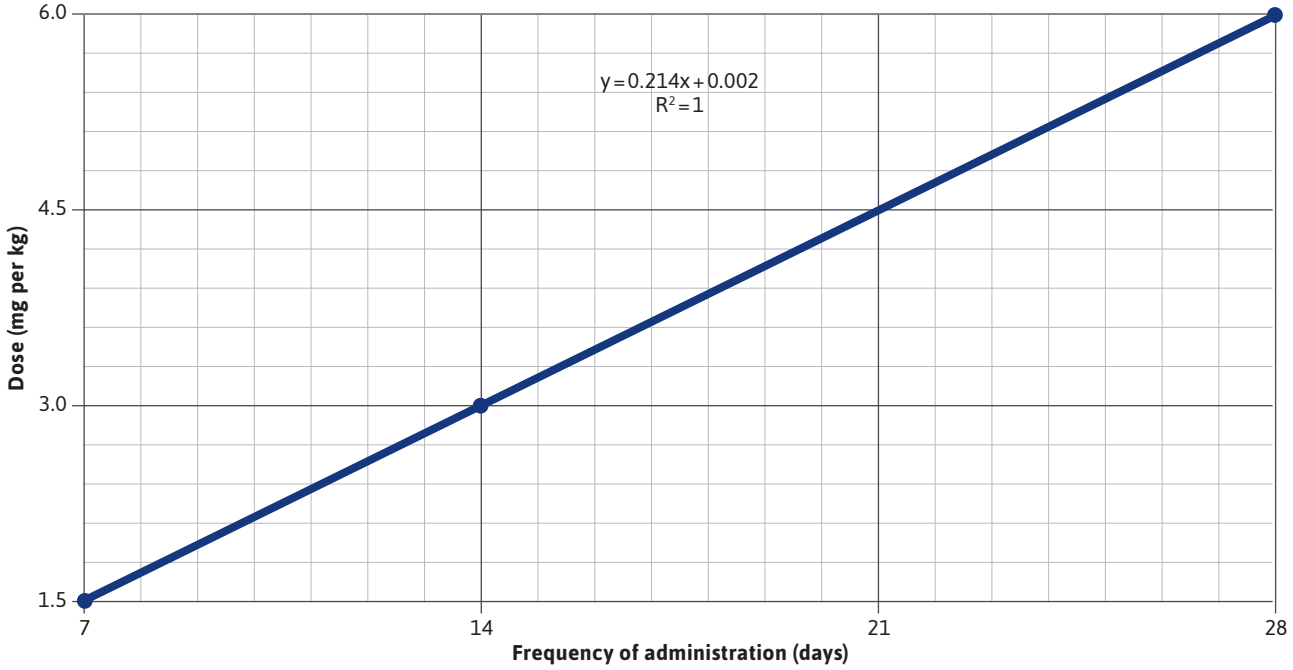
The manufacturer provides a comprehensive Dosing and Administration Guide on their website validating this concept.[10] The guide presents vial combination options for doses from 9 to 900 mg per dose; vial combinations for single doses greater than 90 mg are accomplished with a maximum of 14 mg of waste per this tool. In addition to the dosing guide, the manufacturer also provides the Hemlibra Dosing Calculator.[11] Health care professionals enter the patient weight (kilograms or pounds) and the calculator presents vial combination options for the 3 on-label regimens. Although both tools are innovative, the FDA-approved product labeling directing dose calculations to one-tenth of 1 milligram, limits their capacity to manage medication waste completely. Consider this: doses ranging from 1.45 mg/kg to 1.54 mg/kg all round to a dose of 1.5 mg/kg. Selecting a dose that falls within this range increases the probability that it can be accomplished by using full vials.

Waste may also be managed by leveraging the predictable pharmacokinetics of emicizumab-kxwh following a 4-week loading regimen.[7] Yu et al[12] published a simulation study describing the impact of dose rounding to patient-level pharmacokinetics. They demonstrated that steady-state plasma levels remain within a reference range linked to efficacy even when doses are rounded up or down to the nearest vial. This work built on the pharmacometric approach first applied by Yoneyama et al[7] to select dosing regimens for use in clinical trials in the absence of a conventional dose-finding study.

Plotting the 3 approved maintenance regimens as dose (mg/kg) vs frequency (days) reveals that for every 1 day the interval is adjusted, the dose should change by plus or minus 0.2143 mg/kg ($y = 0.2143x$; $R^2 = 1$) to maintain expected plasma steady state and associated efficacy (Figure 1).

The linear pharmacokinetics of emicizumab-kxwh support custom regimens outside the standard weekly,

FIGURE 1 Maintenance Dose (mg/kg) vs Frequency of Administration (Days)



every 2- or every 4-week intervals directed in the product labeling (eg, 2.14 mg/kg every 10 days). Leveraging these options can reduce injection frequency, potentially improving adherence and quality of life. Custom frequencies require interpretation and application of data obtained from published literature that cannot be integrated into the manufacturer dosing tools due to FDA rules.[7,12,13] The pharmacist can apply these concepts, offering patient-level recommendations for consideration by the prescriber to optimize the regimen and minimize discarded drug.

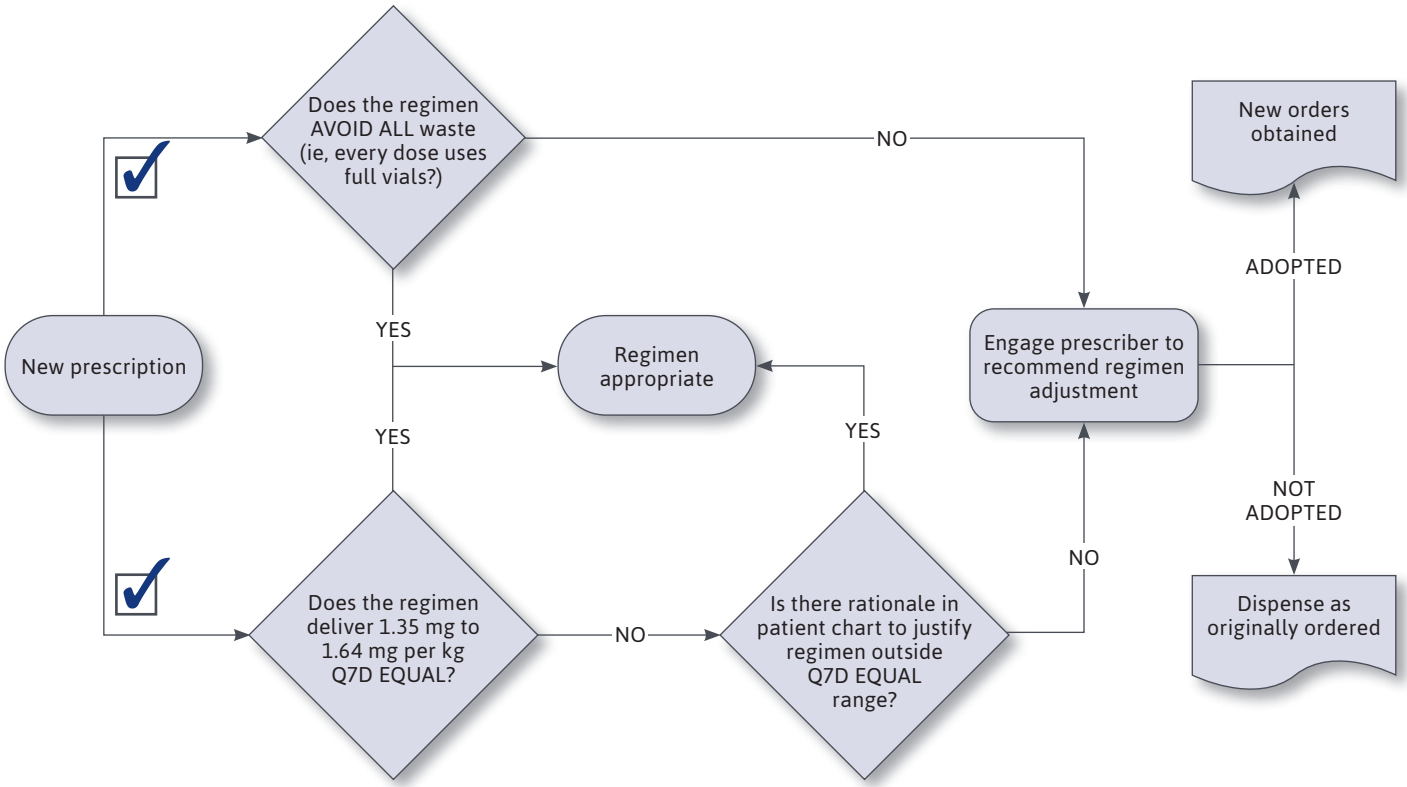
Recent treatment guidelines from the National Hemophilia Foundation-Medical and Scientific Advisory Council recommend consideration be given to the use of emicizumab-kxwh in "persons with hemophilia A, of any age or severity, with or without inhibitors."[14] This guideline has driven an increase in the number of patients with hemophilia A who are offered emicizumab-kxwh. Increased use in patients without an inhibitor has increased the overall cost of providing prophylaxis.[15]

The Regimen Optimization Algorithm: emicizumab-kxwh (Figure 2) was created to address regimen appropriateness and waste by assuring consistent pharmacist review of every new maintenance prescription in patients aged older than 11 years and inhibitor negative. It was designed with 2 goals in mind: validate that the dose and frequency ordered is likely to deliver a therapeutic steady state concentration and reduce or eliminate medication waste.

To evaluate the potential for subtherapeutic or supratherapeutic dosing, the prescribed dosing regimen is converted to an every-7-day equivalent (Q7D EQUAL) to the nearest hundredth of a milligram. For example, a regimen of 3.0 mg/kg every 14 days converts to 1.50 mg/kg Q7D EQUAL. The purpose of the conversion is to improve interrater reliability when multiple pharmacists apply the algorithm at the patient level.

Further action by the pharmacist is required per protocol on identification of any dosing regimen falling outside the established threshold of 1.35 mg/kg to 1.64 mg/kg Q7D EQUAL. This range was intentionally set at ±10% of the original on-label dose of 1.5 mg/kg every 7 days to identify only the most clinically significant outliers for prescriber follow-up. The range also accommodates insignificant changes in patient weight over time. Appropriateness of this range is supported by guidance that "administered doses within 20% of the target" did not require medical intervention in the phase 3 trial investigating emicizumab-kxwh.[16] Doses lower than this threshold carry the potential to deliver plasma concentrations that are subtherapeutic,

FIGURE 2 Regimen Optimization Algorithm: Emicizumab-kxwh



Q7D = every 7 days.

leaving the patient vulnerable to inadequate bleed control.[17] Doses higher than this threshold carry the potential to deliver plasma concentrations that are supratherapeutic, a class (antibody) phenomenon sometimes associated with toxicity.[13]

Action involves pharmacist review of the medical record for clinical rationale to justify the dosing outlier. In the absence of documented rationale, prescriber engagement is required to either obtain one or recommend a dose adjustment. Prescriber-defined rationale for lower-than-threshold dosing includes hyperalbuminemia, severe hepatic impairment, side-effect intolerance, or cardiac history favoring some level of anticoagulation.[14,18] Prescriber-defined rationale for higher than threshold dosing is limited to a history of breakthrough bleeding.

Once the mg/kg Q7D EQUAL dose is deemed appropriate, the regimen is then assessed for waste in the form of partial vial use. Further action by the pharmacist is required per protocol on identification of partial vial use and involves engaging the prescriber to recommend an alternate vial combination.

Evidence-based talking points in support of recommendations are offered as appropriate to the situation and include (1) accuracy of dosing to the 10th of 1 mg is not feasible when using the 2-mL or 3-mL syringe recommended in the product labeling for doses greater than 150 mg as they are marked in increments of 0.2 mL,[19] (2) actual body weight can fluctuate up to 1-2 kg day to day,[20] (3) decreased potential for dosing errors when patients are instructed to draw up full vial(s) for self-injection,[21] (4) following the loading regimen, linear pharmacokinetics and dose proportionality of emicizumab-kxwh is established up to 50 mg/kg per dose,[16] (5) literature validation in support of rounding most emicizumab-kxwh doses to the nearest vial(s) to avoid or eliminate waste and decrease cost without negatively impacting outcomes is published,[12] and (6) reducing injection burden by extending the dosing interval requires fewer

injections over time, carrying the potential to improve adherence and overall quality of life.[22]

Prescriber-adopted recommendations become new prescriptions dispensed by the pharmacy. When prescribers decline to adopt the pharmacist recommendation, rationale is obtained and documented in the patient medical record.

Pharmacist application of this algorithm is required and described in the specialty pharmacy standard operating procedure for dispensing emicizumab-kxwh. It is applied to all patients regardless of payer (ie, commercial, government, or health plan). We present the results of this program, shown over 1 full year, to optimize the emicizumab-kxwh regimen and minimize waste.

Methods

A retrospective review of electronic medical records for 360 patients receiving emicizumab-kxwh as maintenance therapy to prevent bleeding episodes in hemophilia A was conducted. Patients' data were included in the final sample if they were male, were aged 18 to 89 years, were inhibitor negative, and had at least 1 new or renewal prescription filled during the study period of January 1, 2021, to December 31, 2021. Three hundred seventy regimens were identified as evaluated per protocol and included in the final sample. Baseline data included patient age, dosing weight, the emicizumab-kxwh dose and interval originally prescribed, the every-7-day equivalent (Q7D EQUAL) regimen (to the hundredth of a milligram), the vial combination ordered per dose, and the resulting waste, if any, generated per dose.

Each regimen was reviewed for pass/fail status. Failing regimens were further reviewed for the underlying reason(s) driving the failure: partial vial use, dose greater than threshold of 1.64 mg/kg Q7D EQUAL, dose less than threshold of 1.35 mg/kg Q7D EQUAL, or combination of partial vial and dose outside threshold. Documentation of prescriber engagement to recommend a regimen adjustment was reviewed and outcome data collected. Data included the new emicizumab-kxwh dose and interval, Q7D EQUAL, vial combination, and waste generated per dose or prescriber rational for declining the recommendation. In cases in which the regimen passed but the prescriber was engaged with intent to improve the regimen, it was considered failed by professional judgment.

For adopted recommendations, the regimen was compared before and after prescriber engagement to quantify waste (in mg) avoided. The average sales price (ASP) per mg was applied to calculate savings.

Results

Of 370 reviewed regimens, 48% (176/370) failed pharmacist clinical review; 163 failed the evidence-based algorithm, whereas 13 were judged suboptimal per professional judgment (ie, more vials or larger volume per dose than necessary). Single reasons for failing the algorithm included partial vial use (80%; 131/163), dose greater than 1.64 mg/kg Q7D EQUAL (11%; 17/163), or dose less than 1.35 mg/kg Q7D EQUAL (5%; 8/163). Seven (4%; 7/163) regimens failed on 2 points; 5 required the use of partial vials and delivered a dose equivalent higher than threshold and 2 required the use of partial vials and delivered a dose equivalent lower than threshold.

For 112 regimens, prescribers were offered recommendations to optimize the emicizumab-kxwh regimen. In a small majority of cases (53%; 59/112), the prescriber declined to adopt the recommendation and the prescription was filled as originally ordered. The sole rationale for not adopting a recommendation to increase the dose to greater than 1.34 mg per Q7D EQUAL was established efficacy. On further review, all 3 patients falling into this cohort realized weight gain and the prescribers declined to recalculate the dose to the updated weight.

The most common prescriber rationale for not adopting a recommendation to decrease the dose to less than 1.64 mg/kg Q7D EQUAL was assessment that baseline adherence was poor enough to justify the higher dose (37%; 4/11). Additional rationale included a patient history of breakthrough bleeding with lower-dose regimens (27%; 3/11), established efficacy in the absence of side effects (18%; 2/11), and dosing to ideal body weight in obesity (9%; 1/11). No prescriber rationale was offered for the remaining case (9%; 1/11).

The most common rationale for requiring the use of partial vials was prescriber refusal to round the dose citing desire to dose precisely to the FDA-approved labeling of 1.5 mg/kg per Q7D equivalent (64%; 28/44). Another common prescriber rationale driving partial vial use illustrates a practical issue related to how emicizumab-kxwh is formulated. The product labeling directs that vials of differing concentration should not be mixed in the same syringe. Doses greater than 30 mg and optimized by using a 30-mg vial will always require a minimum of 2 injections. This is because the 30-mg vial has a different concentration (30 mg/mL) than the other 3 (150 mg/mL). It is not uncommon for prescribers to order a 105-mg vial (vs a 30-mg plus a 60-mg vial) to accommodate doses between 61 mg and 89 mg to limit the patient to a single injection per dose. For a weekly regimen of 61 mg, the resulting waste of using a 105-mg vial is 44 mg per dose. At an ASP of $99.808 per mg,

TABLE 1 Patient-Level Detail of Prescriber-Adopted Recommendations

Dose (mg)	Weight (kg)	mg/kg	Days	Q7D Equal	30-mg vial	60-mg vial	10-mg vial	15-mg vial	Dispense per dose (mg)	Waste per dose (mg)	Recommendation and mg/kg/day change (if applicable)		30-mg vial	60-mg vial	105-mg vial	150-mg vial	Dispense per dose (mg)	Waste per dose (mg)	Savings per dose (mg)
Cohort 1: Adjust vial combination (N=11; 23%)																			
120	77.1	1.56	7	1.56	4				120	0	Adjust vial combination	NA		2			120	0	0
960	156.5	6.13	28	1.53		16			960	0	Adjust vial combination	NA		1		6	960	0	0
240	80.8	2.97	14	1.49			1	1	255	15	Adjust vial combination	NA		4			240	0	15
270	97.5	2.77	14	1.38		3	1		285	15	Adjust vial combination	NA		2		1	270	0	15
315	55.8	5.65	28	1.41		2	2		330	15	Adjust vial combination	NA			3		315	0	15
210	72.6	2.89	14	1.45		2	1		225	15	Adjust vial combination	NA			2		210	0	15
90	59.9	1.50	7	1.50			1		105	15	Adjust vial combination	NA	1	1			90	0	15
127	82.1	1.55	7	1.55				1	150	23	Adjust vial combination	NA	1		1		135	8	15
120	81.6	1.47	7	1.47				1	150	30	Adjust vial combination	NA		2			120	0	30
684	113.4	6.03	28	1.51			7		735	51	Adjust vial combination	NA		1	6		690	6	45
210	77.6	2.71	14	1.35				2	300	90	Adjust vial combination	NA			2		210	0	90
Cohort 2: Dose decrease (N=17; 35%)																			
180	108.9	1.65	7	1.65		3			180	0	Dose decrease to 165 mg	0.1377		1	1		165	0	15
300	91.2	3.29	14	1.64				2	300	0	Dose decrease to 270 mg	0.3289		2		1	270	0	30
390	61.7	6.32	30	1.47		4		1	390	0	Dose decrease to 375 mg	0.2431		2	1	1	375	0	15
465	143.8	3.23	14	1.62		1	1	2	465	0	Dose decrease to 450 mg	0.1043				3	450	0	15
555	85.7	6.48	28	1.62			1	3	555	0	Dose decrease to 525 mg	0.3501			5		525	0	30
306	101.9	3.00	14	1.50			3		315	9	Dose decrease to 300 mg	0.0589				2	300	0	15
245	79.4	3.09	14	1.54			1	1	255	10	Dose decrease to 240 mg	0.0630		4			240	0	15
184.5	122.5	1.51	7	1.51	1	1	2		195	10.5	Dose decrease to 180 mg	0.0367		3			180	0	15
544	90.7	6.00	28	1.50			1	3	555	11	Dose decrease to 540 mg	0.0441		4		2	540	0	15
123	85.3	1.44	7	1.44	1		1		135	12	Dose decrease to 120 mg	0.0352		2			120	0	15
213	72.6	2.93	14	1.47		2	1		225	12	Dose decrease to 210 mg	0.0413			2		210	0	15
243	81.6	2.98	14	1.49			1	1	255	12	Dose decrease to 240 mg	0.0368		4			240	0	15
212.7	140.6	1.51	7	1.51		2	1		225	12.3	Dose decrease to 210 mg	0.0192			2		210	0	15
242	80.3	3.01	14	1.51			1	1	255	13	Dose decrease to 240 mg	0.0249		4			240	0	15
315	85	3.71	14	1.85	1			2	330	15	Dose decrease to 255 mg	0.7059			1	1	255	0	75
345	102.1	3.38	14	1.69		1		2	360	15	Dose decrease to 300 mg	0.4407				2	300	0	60
486	83.9	5.79	28	1.45		1		3	510	24	Dose decrease to 480 mg	0.0715	1			3	480	0	30

continued on next page

TABLE 1 **Patient-Level Detail of Prescriber-Adopted Recommendations** *(continued)*

Dose (mg)	Weight (kg)	mg/kg	Days	Q7D Equal	30-mg vial	60-mg vial	10-mg vial	15-mg vial	Dispense per dose (mg)	Waste per dose (mg)	Recommendation and mg/kg/day change (if applicable)		30-mg vial	60-mg vial	105-mg vial	150-mg vial	Dispense per dose (mg)	Waste per dose (mg)	Savings per dose (mg)
Cohort 3: Dose increase (N = 18; 38%)																			
105	79.4	1.32	7	1.32			1		105	0	Dose increase to 120 mg	–0.1889		2			120	0	–15
300	112	2.68	14	1.34				2	300	0	Dose increase to 345 mg	–0.4018		4	1		345	0	–45
164	105	1.56	7	1.56		1	1		165	1	Dose increase to 165 mg	–0.0095		1	1		165	0	0
103.5	68.9	1.50	7	1.50			1		105	1.5	Dose increase to 105 mg	–0.0218			1		105	0	0
313.5	104.3	3.01	14	1.50			3		315	1.5	Dose increase to 315 mg	–0.0144			3		315	0	0
147	97.5	1.51	7	1.51				1	150	3	Dose increase to 150 mg	–0.0308				1	150	0	0
222	147	1.51	7	1.51		2	1		225	3	Dose increase to 225 mg	–0.0204		2	1		225	0	0
267	90.7	2.94	14	1.47		2		1	270	3	Dose increase to 270 mg	–0.0331		2		1	270	0	0
342	111.1	3.08	14	1.54		4	1		345	3	Dose increase to 345 mg	–0.0270		4	1		345	0	0
235.5	79.2	2.97	14	1.49		4			240	4.5	Dose increase to 240 mg	–0.0568		4			240	0	0
145	93	1.56	7	1.56				1	150	5	Dose increase to 150 mg	–0.0538				1	150	0	0
280	94.8	2.95	14	1.48		3	1		285	5	Dose increase to 300 mg	–0.2110				2	300	0	–15
144	102.1	1.41	7	1.41				1	150	6	Dose increase to 150 mg	–0.0588				1	150	0	0
112.5	79.4	1.42	7	1.42		2			120	7.5	Dose increase to 120 mg	–0.0945		2			120	0	0
290	95.5	3.04	14	1.52				2	300	10	Dose increase to 300 mg	–0.1047				2	300	0	0
287	98.9	2.90	14	1.45				2	300	13	Dose increase to 300 mg	–0.1314				2	300	0	0
135	103.6	1.30	7	1.30				1	150	15	Dose increase to 150 mg	–0.1448				1	150	0	0
216	74.8	2.89	14	1.44		4			240	24	Dose increase to 225 mg	–0.1203		2	1		225	0	15
Cohort 4: Dose increase with interval adjustment (N = 2; 4%)																			
129.5	86.3	1.50	7	1.50	1		1		135	5.55	Dose increase to 225 mg; interval to 14 days	NA		2	1		225	0	NA
194	63.5	3.06	14	1.53		1		1	210	16	Dose increase to 210 mg; interval to 15 days	NA			2		210	0	NA

Each cohort is grouped by recommendation offered; sorted lowest to highest waste per dose (original prescription).
NA = not applicable; Q7D = every 7 days.

the cost of this waste per 28 days is $17,566.21 (annum of $228,360.70).[23] In this dataset, 6 (14%; 6/44) recommendations were declined citing prescriber preference limiting each dose to 1 injection. Additional prescriber rationale for partial vial use included assessment by the prescriber that a custom frequency was impractical at the patient level (4.5%; 2/44), anticipated weight gain requiring titration between refills to keep the Q7D EQUAL dose greater than 1.34 mg/kg (4.5%; 2/44), established efficacy at current dose (2%; 1/44), and a desire to use fewer vials per dose (2%; 1/44). No prescriber rationale was offered for the remaining 9% (4/11) of cases.

For 48 of 112 (43%) regimens, the prescriber adopted the pharmacist recommendation (the remaining 4 cases [4%; 4/112], the prescriber took the recommendation under advisement until the next clinic appointment). Table 1 provides patient-level detail describing the emicizumab-kxwh regimen before and after pharmacist intervention, including any waste (in mg per dose) generated.

For purposes of analysis, waste is defined as the amount of drug that is discarded with each dose. For example, use

TABLE 2 Cumulative Waste and Savings by Cohort

Cohort	Dose (mg) (original order)	Waste (mg) (original order)	Recommendation	Dose (mg) (dispensed)	Waste (mg) (dispensed)	Savings (ASP) ($)
1	3,615	269.00	Adjust vial combination	3,360	14	56,890.56
2	5,505	155.80	Dose decrease	5,100	0	83,838.72
3	3,915	106.00	Dose increase	3,975	0	(14,971.20)
4	345	21.55	Other	435	0	50,902.08

ASP = average sales price.

of a 150-mg vial of emicizumab-kxwh to accommodate a 147-mg dose would result in 3 mg of waste per dose. Waste is eliminated when the entire contents of the vial or vials are injected by the patient to provide a therapeutic benefit. Avoidance of medication waste has long been a cultural cornerstone unique to the management of hemophilia owing to the plasma-derived nature of traditional clotting factor concentrates, which relied on the altruism of a finite number of plasma donors. In the example cited above, the 3 mg of waste generated per dose could be eliminated by rounding the dose up to 150 mg.

Savings is defined as less drug dispensed per dose. For example, use of a 150-mg vial of emicizumab-kxwh to accommodate a 120-mg dose would result in 30 mg of waste per dose. This 30 mg could be saved by using two 60-mg vials instead of one 150-mg vial.

Table 2 provides cumulative savings by cohort. Cohort 1 included 11 regimens (11/48; 23%), for which the recommendation to adjust the combination of vials dispensed to achieve the originally prescribed emicizumab-kxwh dose was adopted. Combining alternate vials while achieving the same dose reduced overall waste from 269 mg to 14 mg and saved 255 mg (3,615 mg – 3,360 mg) per dose across these 11 patients. When factoring in the number of doses per dispense, driven by frequency of administration at the patient level, the total monthly savings to payers in this group was $56,890.56 (570 mg).

Cohort 2 included 17 regimens (17/48; 35%), for which the recommendation to decrease the emicizumab-kxwh dose, all accomplished by dispensing fewer mg per dose via smaller vials or an alternate vial combination, was adopted. Decreasing the dose brought the mg/kg Q7D EQUAL below the algorithm threshold of 1.64 mg/kg in 3 cases, decreasing the potential for adverse drug reactions associated with supratherapeutic dosing. For 11 regimens, the dose decrease was accomplished by reducing the daily mg/kg dose by less than 0.105 mg/kg (0.0192-0.104), significantly lower than the 0.214-mg/kg change for which a frequency reduction of 1 day could be applied (Figure 1). The final 3 regimens required a dose reduction of greater than 0.214 mg/kg per day (0.243 mg/kg, 0.329 mg/kg, 0.350 mg/kg); in each case, the physician did not adopt the recommendation to also shorten the frequency of administration. These recommendations avoided all 155.8 mg of waste and saved 405 mg (5,505 mg – 5,100 mg) per dose across these 17 patients. When factoring in the number of doses per dispense, driven by frequency of administration at the patient level, the total monthly savings to payers in this group was $83.838.72 (840 mg).

Dose reduction in any context drives concern around potential loss of efficacy. To explore this, the medical records of the 17 patients for whom the dose was decreased were reviewed for patient-reported breakthrough bleeding or administration of clotting FVIII. For 12 patients, the intervention occurred with the first maintenance prescription; consequently, there exist no data to compare bleed frequency and use of clotting FVIII before and after the dose decrease. Data were available for 5 patients who were on the original dose for between 1 and 20 months prior to the dose reduction. There was no change in spontaneous bleed frequency or use of factor to manage trauma/injury following the dose reduction (Table 3).

Cohort 3 included 18 regimens (18/48; 38%), for which the recommendation to increase the emicizumab-kxwh dose was adopted. Increasing the dose brought the mg/kg Q7D EQUAL above the algorithm threshold of 1.34 mg/kg in 3 cases, decreasing the potential for breakthrough bleeding associated with subtherapeutic dosing. In 13 of the cases, the dose was rounded up to match vials already being dispensed, driving no additional savings but avoiding all waste. These dose increases were accomplished by increasing the mg/kg dose by less than 0.132 (0.0095-0.131), significantly lower than the 0.214 mg/kg change for which a frequency extension of 1 day could be applied (Figure 1). For the other 2 cases in this cohort, the dose increase was combined with a vial combination adjustment; one drove the dispense

TABLE 3 Patient-Reported Bleeding/Clotting Factor Concentrate Use Before/After Emicizumab-kxwh Dose Reduction

Original dose (mg)	mg/kg Q7D EQUAL	Months on original	Spontaneous bleeding events	Induced bleeding events	Dose reduction (mg)	Dose dispensed (mg)	mg/kg/day change	Spontaneous bleeding events	Induced bleeding events
123	1.44	1	0	0	3	120	0.0352	0	0
213	1.47	11	0	0	3	210	0.0413	0	0
242	1.51	20	0	1 (leg surgery) 1 (oral) 3 (PT)	2	240	0.0249	0	1 (PT)
300	1.64	12	12	1 (injury) 3 (surgery)	30	270	0.3289	1	0
486	1.45	6	0	0	6	480	0.0715	0	0

PT = physical therapy; Q7D = every 7 days.

amount up by 15 mg and one drove the dispense amount down by 15 mg for a net change across this cohort of zero. These recommendations resulted in reducing all 106 mg of waste across these 18 patients. The overall deficit to this cohort was capped at 60 mg per dose, which was much lower than expected for 18 dose increases. When factoring in the number of doses per dispense, driven by frequency of administration at the patient level, the total monthly cost increase to payers in this group was $14,971.20 (150 mg).

Dose increase in any context drives concern about side effects. To explore this, the medical records of the 18 patients for whom the dose was increased during the study period were reviewed for patient-reported side effects. As part of the standard operating procedure, patient contact is documented with every dispense and includes responses to proactive screening questions designed to identify barriers to care, including side effects. Average follow-up for the 18 patients was 9.7 months (range 5-15). There was 1 report of cough, 1 of hypoglycemia, and 1 myocardial infarction. All 3 were independently adjudicated by the prescriber in real time and deemed unrelated to the administration of emicizumab-kxwh. The cough was related to seasonal allergies, the hypoglycemic event triggered a new diagnosis of diabetes, and the myocardial infarction occurred in a patient who was in a high-risk group (79-year-old male on triple therapy for hypertension). In all cases, emicizumab-kxwh was continued.

The final 2 patients (cohort 4: 2/48; 4%) did not fit in any of the 3 more general cohorts. In case 1, the dose was increased in combination with an interval extension to every 14 days. This regimen change decreased overall administration burden to the patient through 26 fewer injections per year, avoided 5.55 mg of waste per dose, and saved 90 mg per month through vial mix adjustment (total savings to the payer of $8,982.72). In case 2, the dose was increased in combination with an interval extension to every 15 days. This regimen change decreased overall administration burden to the patient through 2 fewer injections per year, avoided 16 mg of waste per dose, and saved $41,919.36 to the payer through one fewer dispense per year (dispensing 12 × 30 days supply vs 13 × 28 days supply).

Discussion

Cumulative waste avoided across all patients was 538.35 mg per dose. Savings of 600 mg per dose was realized across 46 patients (cohorts 1-3). When factoring in the number of doses per dispense, the total monthly savings for cohorts 1-3 was $125,758.08. Addition of the data from the first case in cohort 4 brought monthly savings to $134,740.80.

Total annual payer savings, including the savings driven through 1 fewer dispense illustrated through the second case in cohort 4, was $1,793,549.76. This level of savings is in the context of increasing doses for 42% (20/48) of patients in the sample.

LIMITATIONS

There are multiple limitations to this retrospective review and its applicability to general pharmacy practice. Exclusion of patients aged 12-17 years decreased the sample size by approximately 25%, underestimating the overall savings achieved. Patients of this age range are also more likely to have a dosing weight of less than 60 kg (and dose of 90 mg), skewing data around prescriber rationale for declining

recommendations to reduce medication waste. Additionally, savings is more difficult to precisely determine when more than 1 adjustment is made to a particular regimen (ie, dose and frequency of administration adjustment or frequency of administration and vial combination adjustment).

Patient-level adherence was not formally validated in this study. Rather, it was assumed based on consecutive days of service for each dispense. Waste associated with nonadherence to the medication is therefore unaccounted.

Finally, application of this protocol required dedicated subspecialist pharmacists, trained in the management of hemophilia, sufficient to manage approximately 70 patients each. This level of subspecialization could limit the practicality of launching such a program more globally.

Conclusions

There is evidence to suggest that minor adjustments to dose or frequency can be made to the emicizumab-kxwh regimen to reduce or avoid medication waste. At an ASP of $99.808 per mg, even small reductions in waste can lead to significant savings to a payer, and to the patient in the context of copays or coinsurance, over the course of a full year. The cumulative savings achieved over the 48 regimens, for which the prescriber adopted the specialist pharmacist recommendation totaled $1,793,549.76. This savings was realized despite increasing the dose in close to half of cases. It was also realized in the face of unchanged efficacy and tolerance to therapy.

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DISCLOSURES

Accredo, Inc, a specialty pharmacy dispensing emicizumab-kxwh to patients with hemophilia A, employs the authors. Accredo, Inc, is a subsidiary of EverNorth, which also includes Express Scripts, a PBM that adjudicates a portion of emicizumab-kxwh claims.

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